Issues with First Quantization Quantum Mechanics on Curved Space-Times

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Abstract

We begin with the Bohr hydrogen atom where the electron is saved from infall, due to electromagnetic radiation, by the imposition of a quantum mechanical wave equation, namely the Schrödinger equation. An analogy will then be made to the classical circular orbit of light around a massive body in the Newtonian gravitational potential, where an energy extraction mechanism exists in the form of gravitational radiation and, therefore, a finite lifetime for the photon orbit. We derive a quantum mechanical wave equation suitable for the photon and demonstrate that there are no bound states for the $1/r$ gravitational potential due to scale invariance of our operator, and that the imposition of a ground state yields non-normalizable solutions. The potential-free formulation is then presented, wherein our photon wave equation is put into fully covariant form and is solved on a Schwarzschild background. Techniques from the spectral theory are used to demonstrate that even here there exist no bound states. In the hopes that we can save bound states by re-inserting a mass for our particle, in analogy to Dirac hydrogen, we investigate the fully covariant Dirac equation on curved space-times but find that our problem with a lack of bound states remains. It is then demonstrated that there is reason to believe that the absence of bound states is due to non-normalizability of the wave-function in the presence of an event horizon.

Dedication

פאר די ליבע פון מיין משפחה און באלדער.

Introduction

The motivating idea behind the research that became this thesis arose in the first week of an introductory course in General Relativity. The professor — soon to become my thesis advisor — lectured on the deflection of light by a massive body, and derived the classical result for the deflection angle (too small by a half when compared to the properly general relativistic treatment). Thoughts of tidal forces on extended bodies were floating around my mind for some reason, and it occurred to me to ask if a properly quantum mechanical treatment of the photon as a spatially extended wave-packet would change this prediction, as parts of the 'energy' of the photon would be closer to the massive body than others. I was encouraged to look into the formulation of a wave equation to describe single photons, which was the topic of a thesis [10] advised by Professor Franklin the previous year. After a brief survey, a precise research question and tentative plan were posed, wherein the topic of investigation would be the effects of an external gravitational field on the interior energy dynamics of a single photon wave-packet. I dreamed of braids of light, and perhaps even gravitational self interaction a la Wheeler's geons [27].

Naturally, the first step would be coupling the photon wave function Ψ to the Poisson equation for Newtonian gravity, where the sources are replaced with the energy density $h\nu\Psi^*\Psi$ as

$$\nabla^2\Phi = 4\pi G h\nu\Psi^*\Psi. \tag{1}$$

A problem arises in how to approach coupling Φ back into the photon wave equation, and so, to gain a better understanding of the nature of a potential formulation of gravity in the context of the photon wave equation, began the purportedly simple task of finding a bound state for the photon in a $1/r$ potential. This question, and a motivation for it using an analogy to the issues with a purely classical model of Bohr hydrogen, are where this thesis begins. Significant effort was invested in making this thesis as compelling and accessible as possible. As such this thesis will tell a story. The story begins with physics that will hopefully be very familiar, or at least recognizable, to the reader and will end with an investigation of issues that arise when

attempting to describe single particle quantum mechanics on curved space-times.

Chapter 1

Classical and Quantum Photons in Gravitationally Bound Systems

This chapter sets out to make a photo-gravitational analogue to both the issues with Bohr hydrogen and their resolution, so it is fitting to quickly review this simple system. Bohr's model, introduced in 1913, was an early attempt at describing the hydrogen atom. In the ground state of this system a small positively charged nucleus sits at rest while an electron moves in a circular orbit at a distance of $a_0 \approx 0.53 \times 10^{-10}$ meters from the nucleus. While this model was successful in some ways — it predicted the correct emission lines for hydrogen — it fails in a rather serious manner. Due to the electron's charge and acceleration, a classical analysis suggests that it will continuously radiate energy (as given by the Larmor power law), and thereby the radius will shrink with time, leading to a finite lifetime of $\approx 10^{-11}$ seconds; that is, hydrogen is unstable in the Bohr model. Moving from a classical picture, with the electron orbiting in a Coloumb potential via Newtonian dynamics, to a quantum mechanical one in which the dynamics are governed by the Schrödinger equation, ends up providing a ground state and saving the stable hydrogen atom.

Newtonian gravity allows circular orbits for light, in this setting nothing more than a particle moving at speed c around a massive body at a radius $R = \frac{GM}{c^2}$. We can demonstrate this rather easily with the pseudo-Hamiltonian[1] formulation for our system of interest:

$$H = \frac{1}{2}(\dot{r}^2 + r^2\dot{\phi}^2) - \frac{GM}{r}. \tag{1.1}$$

[1] As we are dealing with a massless particle, this is not really a Hamiltonian, lacking the m that would provide the proper units, but this treatment will serve well for our purposes.

Using the ϕ-component of our equation of motion

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\phi}} = \frac{d}{dt}\left(r^2\dot{\phi}\right) = 0 \tag{1.2}$$

we see that we have a conserved quantity, and so we substitute a constant for $\dot{\phi} = \frac{J_z}{r^2}$, where J_z is the only non-zero component of our pseudo-angular momentum. So now we have

$$H = \frac{1}{2}\dot{r}^2 + \frac{1}{2}\frac{J_z^2}{r^2} - \frac{GM}{r}, \tag{1.3}$$

and we can identify the last two terms as an effective potential

$$U_{\text{eff}}(r) = \frac{1}{2}\frac{J_z^2}{r^2} - \frac{GM}{r}. \tag{1.4}$$

We can determine the minimum of this effective potential by setting $\frac{\partial U_{\text{eff}}}{\partial r} = 0$ to get

$$-\frac{J_z^2}{r^3} + \frac{GM}{r^2} = 0 \tag{1.5}$$

which gives us $r_{min} = \frac{J_z^2}{GM} = \frac{\dot{\phi}^2 r^4}{GM}$. Setting $\dot{\phi} = \frac{c}{r}$, appropriate for light, we find that our circular orbit occurs at a radius $R = \frac{GM}{c^2}$.[2]

Now we have a circular orbit for a photon around a massive body, but if we desire to make the continued analogy to Bohr hydrogen then we need a method of energy extraction. While the Larmor power law $P = \frac{q^2 a^2}{6\pi\epsilon_0 c^3}$ seems to apply to charged particles only, and light is certainly neutral, we can make the weak-field analogy of general relativity where gravity is treated in the same form as the theory of electromagnetism, albeit with only one type of charge (i.e. no negative mass). That is, classical pre-Einsteinian gravity is a scalar theory and differs from the electric potential only in its coupling constants and absence of negative charge. In this setting, one sends $\frac{q^2}{4\pi\epsilon_0} \longrightarrow -Gm^2$ to recover the analogous prediction for the radiation produced by an accelerating massive particle. However, we can no longer interpret the radiation as electromagnetic (as there are no moving charges to act as a source) and must instead accept that there is some form of gravitational radiation, or gravity waves, carrying energy away from the system. The existence of gravitational waves in the fully general relativistic theory will be taken up later in this chapter.

With our electromagnetic-gravitational replacement we arrive at an expression for

[2] Half of the Schwarzschild radius!

the energy carried away via gravitational waves

$$P = -\frac{2}{3}\frac{Gm^2a^2}{c^3}. \tag{1.6}$$

Which, for our photon orbit, becomes

$$P = -\frac{2}{3}\frac{m^2c^5}{GM^2}. \tag{1.7}$$

We are now faced with the issue of identifying an effective rest mass for our photon, as it seems one is needed to quantify the energy carried away by gravitational radiation. We utilize the Compton wavelength, wherein $m_{\text{eff}} = \frac{h\nu}{c^2}$ where ν is the frequency of light. Finally, we have:

$$P = -\frac{2}{3}\frac{(h\nu)^2c}{GM^2}. \tag{1.8}$$

The negative sign appearing above in our gravitational power radiated would seem to suggest that our system is actually *absorbing* energy through gravitational radiation. This is certainly nonsensical and aphysical: where would this energy be coming from? Also, in continuing the analogy to electromagnetic Larmor case, we expect the radiation to carry energy *away* from the system. So, where did our negative sign come from? It came from our introduction of a minus sign in the transcription from electromagnetic constants to gravitational ones, in order to assure a purely attractive Newtonian gravitational potential. It seems that the introduction of this negative sign, while certainly correct, ends up introducing a sign ambiguity when carrying out the Larmor power argument. So, we will ignore the aphysical part and take only the magnitude of this quantity seriously. The reader will see in Section 1.5 that a more rigorous derivation of the energy loss via gravitational radiation in the general relativistic setting has the proper sign.

1.1 Classical Infall Time

In what follows, we calculate the infall time in the non-relativistic regime, in a treatment exactly paralleling that of Bohr hydrogen. While the photon is certainly relativistic, and as such we don't expect to get out proper numerical results, this treatment will serve to demonstrate the general time scales involved in our problem of interest, before we proceed to try and save its stability with a quantum mechanical wave equation.

We seek to re-write Eq. 1.6 in terms of the radial variable r so that we may build a

differential equation in r to find the infall time. The non-relativistic force law $F = ma$ tells us that $\frac{GMm}{r^2} = ma$ and so we can rewrite Eq. 1.6 as

$$\frac{dU}{dt} = -\frac{2}{3}\frac{G^3m^2M^2}{r^4c^3}. \tag{1.9}$$

The total non-relativistic energy is

$$U = -\frac{GMm}{r} + \frac{1}{2}mv^2 = -\frac{GMm}{2r}. \tag{1.10}$$

Equating the time-derivative of our total energy to the energy carried away by gravitational radiation we have

$$-\frac{2}{3}\frac{G^3m^2M^2}{r^4c^3} = \frac{GMm}{2r^2}\dot{r}, \tag{1.11}$$

and we can see that

$$-\frac{4}{3}\frac{G^2Mm}{c^3} = r^2\dot{r} = \frac{1}{3}\frac{dr^3}{dt}. \tag{1.12}$$

So, with a bit of integration we arrive at our equation of interest

$$r^3 = r_0^3 - \frac{4G^2mM}{c^3}t. \tag{1.13}$$

This expression gives us an infall time to the origin of

$$t_{\text{fall}} = \frac{r_0^3c^3}{4G^2mM}. \tag{1.14}$$

Plugging in the constants relevant to our problem of interest, that is, $r_0 = \frac{GM}{c^2}$ and $m = \frac{h\nu}{c^2}$ we get

$$t_{\text{fall}} = \frac{GM^2}{4h\nu c}. \tag{1.15}$$

For a stellar mass black hole, and blue light, this gives a lifetime of 10^{60} seconds. This is substantially longer than the age of the universe, by a factor of 10^{43} seconds, but has a timescale comparable to that of the evaporation of a black hole via Hawking radiation and so one could still consider it physically relevant.

However, in the case of the photon there is an ambiguity as to what the physical effects of energy loss due to gravitational radiation actually are for our system of interest. In the above we naively followed the treatment of a non-relativistic particle that is allowed to have any velocity, and in doing so constrained ourselves to considering a physical scenario in which infall occurs. Whereas the electron's only

reasonable mechanism for energy loss is through a decreasing effective orbital velocity, the photon gives us another option, redshift. Moreover, special relativity constrains the photon to travel at speed c locally, and therefore seems to preclude the infall scenario. From this perspective it seems more appropriate to treat our photon as losing energy through reddening, while staying in its orbit at $\frac{GM}{c^2}$. It then fits to treat Eq. 1.8 as a differential equation in energy $E = h\nu$. We have

$$\frac{dE}{dt} = -\frac{2}{3}\frac{E^2 c}{GM^2}. \tag{1.16}$$

which admits a solution of the form

$$E(t) = \frac{E_0}{At + 1}, \tag{1.17}$$

with $A = \frac{2}{3}\frac{E_0 c}{GM^2}$, and we see a natural time constant $\tau = \frac{1}{A}$ whereby the photon has lost half of its initial energy, thus providing a natural way to define the effective photon lifetime. Here again, we can treat the case of blue light in orbit around a one solar mass black hole to find the effective half-life of $\tau \approx 10^{40}$ seconds. In this setting the photon has no real *finite* lifetime, as it can effectively redden all the way down the energy scale but never reaches $E = 0$. Another reasonable metric by which to judge the lifetime would be when the wavelength of the light becomes larger than our system of interest at $\lambda = \frac{GM}{c^2}$.

Whatever the physical scenario, the fact remains that our photo-gravitational system has all the important characteristics of Bohr hydrogen: A circular orbit exists in a $1/r$ potential, there is a mechanism for energy extraction, and a lifetime for our system. It seems plausible then that we can similarly save the stability of our photon orbit system via the introduction of a quantum mechanical wave equation. However, the Schrödinger equation applies to massive, non-relativistic particles, a far cry from a photon, and so is inappropriate for our purposes. We can see directly from the form of the Schrödinger equation that it will not be properly relativistic. First, there is an asymmetry in its treatment of temporal and spatial coordinates. The Schrödinger equation is second order in space but first order in time, and as such fails to capture the symmetry or equivalence in these coordinates that we would expect for a properly relativistic wave equation. Moreover, the non-relativistic expression for the kinetic energy is encoded in the form of the equation. In the following section we present a derivation of a quantum mechanical wave equation appropriate for light.

1.2 A Quantum Mechanical Photon Wave Equation

From the classical theory of electromagnetism we are familiar with the identification of light as the propagating vacuum solutions to Maxwell's equations. That is, light "is" a pair of orthogonal electric and magnetic fields oscillating with a certain frequency and propagating in the direction of travel. Explictly, the source-free Maxwell equations are:

$$\nabla \cdot \mathbf{E}(\mathbf{r}, t) = 0 \qquad \nabla \cdot \mathbf{B}(\mathbf{r}, t) = 0$$
$$\frac{1}{c^2}\partial_t \mathbf{E}(\mathbf{r}, t) = \nabla \times \mathbf{B}(\mathbf{r}, t) \qquad \partial_t \mathbf{B}(\mathbf{r}, t) = -\nabla \times \mathbf{E}(\mathbf{r}, t).$$

These equations can be put in the form of a single equation which dictates the evolution of the electromagnetic field via introduction of the Riemann-Silberstein vector [23] $\mathbf{F}$, a complex combination of the electric and magnetic field vectors:

$$\mathbf{F} = \mathbf{E} + ic\mathbf{B}. \tag{1.18}$$

Taking the original Maxwell equations and combining them suitably we are left with

$$i\partial_t \mathbf{F}(\mathbf{r}, t) = c\nabla \times \mathbf{F}(\mathbf{r}, t) \tag{1.19}$$
$$\nabla \cdot \mathbf{F}(\mathbf{r}, t) = 0, \tag{1.20}$$

and so we have reduced the number of equations by a factor of two, at the cost of complexifying the field. Moreover, in Eq. 1.19 we see the form of a wave equation. That is, the temporal evolution of our quantity $\mathbf{F}$ is related to its spatial variation. Going forward, Eq. 1.19, and equivalent formulations, will be identified as the photon wave equation. However, we desire a quantum mechanical wave equation and what we have here seems classical in nature (even ignoring the absence of Planck's constant). We can immediately make some identifications that both help motivate the usefulness of the Riemann-Silberstein vector and its associated wave equation, as well as draw analogies to the regular quantum wave mechanics.

One can build bilinear expressions from our complex vector $\mathbf{F}$ that represent important dynamical quantities associated with the electromagnetic field, and look

like quantum-mechanical expectation values:

$$E = \int \mathbf{F}^* \cdot \mathbf{F} d\tau \tag{1.21}$$

$$\mathbf{P} = \frac{1}{2ic} \int \mathbf{F}^* \times \mathbf{F} d\tau, \tag{1.22}$$

where E is the energy and $\mathbf{P}$ the momentum. In addition, one can see that these quantities are invariant under a phase rotation of $\mathbf{F}$, that is a multiplication of $\mathbf{F}$ by $e^{i\alpha}$. Multiplication by such a phase factor results in the duality rotation (Misner and Wheeler [1957]) of the field vectors:

$$\frac{\mathbf{E}'}{\sqrt{\epsilon_0}} = \cos\alpha \frac{\mathbf{E}}{\sqrt{\epsilon_0}} - \sin\alpha \frac{\mathbf{B}}{\sqrt{\mu_0}} \tag{1.23}$$

$$\frac{\mathbf{B}'}{\sqrt{\mu_0}} = \cos\alpha \frac{\mathbf{B}}{\sqrt{\mu_0}} + \sin\alpha \frac{\mathbf{E}}{\sqrt{\epsilon_0}}. \tag{1.24}$$

The phase α becomes fixed when coupling the electromagnetic field back to charges, but a free photon has the same invariance in its wave function as in the standard quantum mechanics of massive particles.

So, now we have a wave equation that has been derived from Maxwell's equations, and an associated wave function $\mathbf{F}$ that behaves in analogy to quantum mechanical wave function Ψ. We can now attempt to demonstrate a more direct correspondence between our photon wave equation and the form of more canonical QM wave equations. To do this we can rewrite Eq. 1.19 using the spin-1 matrices s_x, s_y, and s_z.

$$s_x = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{bmatrix}, s_y = \begin{bmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{bmatrix}, s_z = \begin{bmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \tag{1.25}$$

which, in the form given above, are in the representation such that they will act on the Cartesian vector components of the wave function, not those labeled by the eigenvalues of s_z as is the common basis.

Using these spin matrices we can take our curl in Eq. 1.19 and put it in the more familiar form of a divergence using the conversion rule from vector to matrix notation:

$$\mathbf{a} \times \mathbf{b} = -i(\mathbf{a} \cdot \mathbf{s})\mathbf{b}, \tag{1.26}$$

and so our wave equation becomes:

$$i\hbar\partial_t \mathbf{F}(\mathbf{r},\mathbf{t}) = c(\mathbf{s}\cdot\frac{\hbar}{i}\nabla)\mathbf{F}(\mathbf{r},t) \tag{1.27}$$

with $\hbar$ inserted on both sides to make our equation look more quantum. Regardless, if one is familiar with relativistic QM, it is immediately obvious that we have a Dirac-type wave function. Indeed, we can compare our new wave equation with the Weyl (massless Dirac) equation:

$$i\hbar\partial_t \phi(\mathbf{r},\mathbf{t}) = c(\sigma\cdot\frac{\hbar}{i}\nabla)\phi(\mathbf{r},t). \tag{1.28}$$

We see that they differ only in that the Weyl equation has the Pauli matrices appropriate for spin-1/2 particles instead of the spin-1 matrices necessary for our photon wave equation. However, the divergence-less condition for $\mathbf{F}$ actually constrains the number of degrees-of-freedom down to two. When one considers that the photon has only two allowed spin-states, rather than the three one may expect from its status as a spin-1 particle because $S_z = 0$ is disallowed due to the absence of a rest-frame, it becomes more evident that the Dirac equation is a good fit since it has two spin-states encoded. This correspondence is immensely re-assuring as it connects a key, experimentally verified, result of relativistic QM with our attempt to identify the Riemann-Silberstein vector as the appropriate wave function for light. Moreover, if one performs a full quantum field theoretic treatment of light and then constrains the treatment to a single particle, our photon wave equation appears naturally [16].

With our quantum mechanical wave equation in hand, we can now study its properties further. However, going forward we will study the massless Dirac equation as opposed to the photon wave equation we have just derived. Making this choice changes *only* the number of components of the wave-function from $3 \to 2$, and should not change the spectrum. This choice is made as the Dirac equation is a well-studied system, and we can compare the results obtained with the literature more easily. We will assume that the massless-Dirac equation properly describes the characteristics of the photon we want to study (bound states), and we will still consider our system to be describing a photon.

An obvious first step is to construct the stationary solutions of the wave equation as they will serve as our building blocks for all other solutions. We do this by separating the time variable and solving the resulting eigenvalue problem. The relevant

eigenvalue equation reads

$$c(\sigma \cdot \frac{\hbar}{i}\nabla)\Psi(\mathbf{r}) = E\,\Psi(\mathbf{r}). \tag{1.29}$$

1.3 Scale Invariance

We have a circular photon orbit allowed at $\frac{GM}{c^2}$ from the classical physics, and can now study the quantum mechanical system and hope to find a bound state of our system that corresponds to this photon orbit. We begin by coupling the Newtonian scalar potential to our photon wave equation,

$$\left(c\,\sigma \cdot \frac{\hbar}{i}\nabla - \frac{GMh\nu}{c^2 r}\right)\Psi(r) = E\,\Psi(r), \tag{1.30}$$

where we expect a spectrum for our coupled Hamiltonian that corresponds to the energies of various allowed orbits. However, upon manipulating our equation into a radial form in the usual way [25], we find that the resulting equation lacks an encoded length-scale. The relevant radial equation for stationary states is

$$\begin{bmatrix} \frac{-GMh\nu}{c^2 r} & -\frac{\hbar}{i}\frac{\partial}{\partial r} - \frac{\hbar}{r} \\ \frac{\hbar}{i}\frac{\partial}{\partial r} + \frac{\hbar}{r} & \frac{-GMh\nu}{c^2 r} \end{bmatrix} \begin{bmatrix} \Psi_a(r) \\ \Psi_b(r) \end{bmatrix} = \lambda \begin{bmatrix} \Psi_a(r) \\ \Psi_b(r) \end{bmatrix}. \tag{1.31}$$

Indeed, attempts to solve Eq. 1.30 for the eigenstates resulted in an inability to terminate the series solution in the manner necessary for the state to be normalizable, and hence an empty eigenspectrum. Moreover, our problem is scale invariant (all four terms in our radial Hamiltonian go like $1/r$). So even if one were to find an energy for the ground state, one could easily re-scale it and end with any energy they desire. This issue also arises when trying to describe the interaction of an electron with a point-dipole via the Schrödinger equation [12]. Such a phenomena is certainly aphysical, so we recognize that the lack of a mass term in the photon wave equation, as compared to the Dirac equation, results in there being no admissible bound states for the $1/r$ potential. Should we be surprised by this? Perhaps not. The entire physical system is contrived. Gravity does not obey Newtonian theory, it is more properly treated in the general relativistic setting. Moreover, while the Larmor→Einstein analogue for radiated power helps us understand the phenomenon, a rigorous treatment demands the tensorial character of general relativity.[3] So, in the following sections we make

[3]For instance, while a system with a dipole moment will radiate electromagnetically it is forbidden from doing so in the gravitational setting due to momentum conservation.

another pass at telling our story, hoping that with a proper treatment of the system we may be able to save our bound orbit.

1.4 The General Relativistic Photon Sphere

We begin with a brief study of the geodesics of light in the Schwarzschild geometry. We can determine the radius at which the orbit of a photon around a body of mass M will be circular. With the Schwarzschild metric given by:

$$ds^2 = -\left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 + \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 + r^2(\sin^2\theta d\phi^2 + d\theta^2). \tag{1.32}$$

We can now set $dr = 0$ as we are attempting to find motion constrained to a constant radius and $ds = 0$ because we need light-like intervals. Moreover, the θ coordinate can be ignored so J_z is a constant of the motion and we can pick $\theta = \frac{\pi}{2}$. Then $d\theta = 0$ and we have:

$$\left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 = r^2 d\phi^2, \tag{1.33}$$

which we can rearrange to get

$$\frac{d\phi}{dt} = \frac{c}{r}\sqrt{1 - \frac{R_s}{r}}, \tag{1.34}$$

where R_s is the Schwarzschild radius $\frac{2GM}{c^2}$. Using the geodesic equation, we can write:

$$\left(\frac{d\phi}{dt}\right)^2 = \frac{c^2 R_s}{2r^3}, \tag{1.35}$$

and set this equal to the square of Eq. 1.34; which we can rewrite to get:

$$R_{ps} = \frac{3}{2} R_s, \tag{1.36}$$

with R_{ps} the radius of our photon sphere. It is interesting to note that the radius of the photon sphere predicted here by general relativity is three times larger than the Newtonian result, which seems a bit strange when considering that the classic prediction of the deflection angle of light by a massive body yields a general relativistic prediction only *twice* as large as the classical one.

With a radius for our photon sphere in hand, we can now attempt to quantify the energy lost from our system via gravitational radiation. However, this turns out

to be a rather tricky endeavour as one realizes that it is impossible to localize a concept of energy in the gravitational field. That is, space-time is locally flat and hence at any point in space-time the metric becomes Minkowskian and we would say that there is no field, hence no energy. Indeed, Einstein himself had a notorious double-reversal in his opinion on whether gravitational waves were physical; that is, whether they carried energy. While in 1916 he published a derivation of the quadrupole formula quantifying the energy loss due to gravitational radiation in the linearized general relativistic setting, he later, along with Rosen, became convinced that the vacuum solutions were non-physical and simply mathematical artefacts of his field theory. Later still, Einstein reversed his opinion again, and was convinced by Infeld and Robertson that he was in error. These oscillating views generally reflected the divided opinions of the gravitational-physics community regarding the energy associated with the vacuum solutions. It was not until the introduction of the now infamous 'sticky bead' argument,[4] proposed by none other than Richard Feynman, while attending a conference under a pseudonym, that the large majority of physicists came to agree on the physical reality of the prediction of gravitational radiation. In 1974 the discovery of the Hulse-Taylor system, a binary neutron star system in which measurements of the orbital decay precisely matched the loss of energy via gravitational radiation, finally vindicated the fifty-eight year old prediction and the interpretation of gravitational waves as carrying energy away from their source. During the drafting of this document the first direct detection of a gravitational wave was announced [1], a full century after Einstein first proposed the idea, kickstarting an entirely new field of gravitational wave astronomy and, for the first time, allowing us to probe the universe using tools outside the electromagnetic spectrum; we are, in effect, *hearing* the universe.

While quantifying the energy carried away by gravitational waves has a history of controversy, the technique is now firmly grounded and if we proceed carefully we can define the energy enclosed in localized patches of spacetime for our dynamical system of a circular photon orbit, and then properly treat the energy being carried away by gravitational radiation. As a check on our results, we should expect to achieve a close analogue to the results obtained through the electromagnetic-to-gravitational transformation in Section 1.1, modulo customary constants of 2 or 4 [14].

[4] *The argument goes like this: As a gravitational wave with suitable polarization passes by, it stimulates a sticky bead to vibrate back and forth along a rod normal to its direction of travel. The 'stickiness,' or friction, creates heat; thus demonstrating that energy really is carried by the gravitational wave.*(1957) From the modern perspective, this argument seems trivial enough to be obvious, but was apparently quite powerful to the physicists of the day.

1.5 Energy Loss in Linearized General Relativity

From the Einstein field equations:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu} \tag{1.37}$$

we can expand the equations in powers of $h_{\mu\nu}$ with $g_{\mu\nu} = h_{\mu\nu} + \eta_{\mu\nu}$. Defining

$$\tilde{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h^{\sigma}_{\sigma}$$

we can write

$$\tilde{h}_{\mu\nu,\lambda\lambda} - \tilde{h}_{\mu\lambda,\lambda\nu} - \tilde{h}_{\nu\lambda,\lambda\mu} + \eta_{\mu\nu}\tilde{h}_{\lambda\sigma,\lambda\sigma} = -\frac{16\pi G}{c^4}\tilde{T}_{\mu\nu}, \tag{1.38}$$

where $\tilde{T}_{\mu\nu}$ is a combination of $T_{\mu\nu}$ and the non-linear terms in $h_{\mu\nu}$:

$$\tilde{T}_{\mu\nu} = T_{\mu\nu} + \sum_{k=2}^{\infty} S^{(k)}_{\mu\nu} \tag{1.39}$$

where $S^{(k)}_{\mu\nu}$ involves the product of k instances of $h_{\mu\nu}$ and their derivatives. One can see that the divergence of the left side of Eq. 1.38 vanishes, thus we can determine that

$$\partial^{\nu}\tilde{T}_{\mu\nu,} = 0 \tag{1.40}$$

and so there are a number of associated conservation laws. Of particular interest is the conservation law for energy, which we can put in the form:

$$\frac{dE}{dt} = \oint \tilde{T}^{i}_{0} da_i. \tag{1.41}$$

Eq. 1.41 gives us a way of quantifying the energy loss of a system via gravitational radiation once one has calculated $\tilde{T}_{0i}$ from the expansion of Einstein's field equations. We set out to do this now. The invariance of the theory under arbitrary coordinate transformations implies that one may always choose a coordinate system in which $\partial^{\nu}\tilde{h}_{\mu\nu,} = 0$, so we will do this with the hope of making calculations easier. The surface of integration for Eq. 1.41 will be a large sphere enclosing the system. Given that $h_{\mu\nu}$ and its derivatives will be proportional to $1/r$, only terms with a product of two or more $h_{\mu\nu}$'s (or its derivatives) will count toward calculating Eq. 1.41. Moreover, the terms in the surface integral over $S^{(k)}_{0i}$ for $k \geq 3$ will fall off as $1/r$ or faster as

$r \to \infty$. So we are left with

$$\frac{dE}{dt} = \int_{\partial\Omega} S^{(2)}_{0i} da_i. \tag{1.42}$$

That is, only the second order gravitational stress-energy counts. This second order contribution can be calculated from Eq. 1.38 rather easily. Doing this, and setting all terms proportional to $\tilde{h}_{\mu\nu,\nu} = 0$, as well as all terms proportional to $\tilde{h}_{\mu\nu,\lambda\lambda} = 0$ (as they will yield terms in Eq. 1.42 that go like $1/r$), we get:

$$S^{(2)}_{0i} = \frac{1}{32\pi G}\big[h_{\mu\nu,0}h_{\mu\nu,i} + 2h_{\mu 0,\nu}h_{\mu i,\nu} - 2h_{0\mu,\nu}h_{i\nu,\mu} + 2h_{0i,\mu\nu}h_{\mu\nu} + 2h_{\mu\nu}h_{\mu\nu,0i} - 2h_{\mu\nu}h_{0\mu,\nu i} - 2h_{\mu\nu}h_{i\nu,\mu 0}\big]. \tag{1.43}$$

All of the spatial derivatives above can be re-written as time derivatives multiplied by the appropriate direction vectors n_i as follows

$$\tilde{h}_{\alpha\beta,i} = \frac{\partial}{\partial x_i}\tilde{h}_{\alpha\beta} = -\frac{x_i}{r}\tilde{h}_{\alpha\beta,0} \equiv n_i\tilde{h}_{\alpha\beta,0}.$$

After converting spatial derivatives to temporal, we can consider the time average of these quantities in analogy to the EM treatment. Since we have all time derivatives, we can integrate by parts with impunity. Doing this, and imposing the transverse-traceless gauge condition we end up with

$$\frac{dE}{dt} = -\frac{G}{32\pi c^5}\int_{\partial\Omega} \big\langle \dot{h}_{\alpha\beta}\tilde{h}_{\alpha\beta,i}\big\rangle da_i. \tag{1.44}$$

At this point the canonical method is to project back to the non transverse-traceless gauge, perform the integration, and rewrite our expression in terms of the quadrupole moment q^{jk}. The details are a bit messy, and the algebra will not be carried out here.[5] After all is said and done, we are left with an expression for the energy carried away from a system via gravitational waves in terms of the system's quadrupole moment:

$$\frac{dE}{dt} = -\frac{G}{32\pi c^5}\big\langle \dddot{q}^{jk}\dddot{q}_{jk}\big\rangle. \tag{1.45}$$

One may question the applicability of the derivation just presented to our photon orbit, as in our treatment we dealt with a system of time-varying quadrupole moment with two *massive* bodies. However, there is no concern. This author feels that it is more illuminating to calculate the power loss for the general massive case and then insert the energy-mass equivalence relation for the photon. We will now apply Eq. 1.45

[5] The interested reader is directed to section 7.6 of Carroll's excellent text 'Spacetime and Geometry.'

to our physical system of interest, the circular orbit of light around a Schwarzschild black hole.

1.6 Lifetime of the Photon Sphere

We find the frequency of a circular photon orbit around a Schwarzschild black hole, using the previous result for the radius of the photon sphere:

$$\omega = \frac{c}{R_{ps}} = \frac{c^3}{3GM}. \tag{1.46}$$

We can now parametrize the trajectory of our photon in Cartesian coordinates. This gives an energy density evaluated along the trajectory:

$$T^{00} = h\nu\delta(x' - R\cos(\omega t))\delta(y' - R\sin(\omega t))\delta(z'), \tag{1.47}$$

from which we can directly calculate the quadrupole moment[14]:

$$q^{jk} = 3\int T^{00}(x')x'^j x'^k d\tau' = 3m \begin{bmatrix} R^2\cos^2(\omega t) & R^2\sin(\omega t)\cos(\omega t) & 0 \\ R^2\sin(\omega t)\cos(\omega t) & R^2\ \sin^2(\omega t) & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{1.48}$$

We can plug directly into Eq. 1.45 and use R_{ps} to get

$$\frac{dE}{dt} = -\frac{128}{5}\frac{G}{c^5}m^2\omega^6 R^4 = -\frac{128}{45}\frac{m^2c^5}{GM^2}. \tag{1.49}$$

As a quick aside, we check that our Einsteinian result for radiated power is similar to Eq. 1.7. We see an agreement between the electromagnetic analogue, and full general relativistic result, up to a factor of $\frac{64}{15} \approx 4$.

We have demonstrated that even in the general relativistic setting, we have all the same characteristics of our system needed to make the Bohr hydrogenic analogy. However, we have previously seen that the flat-space-plus-potential formulation for our photon wave equation fails to save our bound orbit, or institute a ground state of any sort. A rescaled radius for our photon orbit and a slightly stronger expression for radiated power certainly won't change this. Can a fully covariant version of our wave equation serve us better? In the next chapter we introduce some tools and results necessary to treat quantum mechanical wave equations on curved space and to better understand the lack of bound states in the Hamiltonian operator for our photon wave

equation.

Chapter 2

Spectral Theory and Quantum Mechanics on Curved Space-Times

Thus far, this story has been about a search for bound states for a quantum mechanical description of a photon around a massive body. In this chapter we hope to gain a deeper understanding of what a bound state for a quantum mechanical system is, and what our expectations should be for them. This is motivated by the realization that in a potential-free formulation (i.e. on curved space-times) what we have is a free particle, and so one could reasonably ask the question: *why should we expect anything that looks like bound states?* In order to answer this question we hope to make the identification between quantum mechanical bound states and classical bound orbits (or time-periodic solutions), as these types of solutions certainly arise in the general relativistic setting. While this identification was key to our classical-quantum analogies in the previous chapter, it lacked significant rigour and the argument is even weaker in a potential-free setting. To help make this identification tighter we introduce a few new tools for analysing our Hamiltonian operators and will, along the way, provide a new way of understanding the lack of bound states for our photon wave equation in the flat-space-plus-potential setting.

2.1 Spectral Methods

The spectral theory has a long and complex history, having its origins outside of physics, but has proved to be uniquely applicable to many physical problems of interest, including predicting features of atomic spectra. This section is not meant as an overview of the field, and will certainly fail to treat many interesting problems, but is instead included to introduce a few concepts and theorems that will prove necessary to

examining our physical problem of interest; namely, the Hamiltonian operator for our photon wave equation. As such, this treatment will certainly lack some of the rigour expected from a mathematician, and the discerning reader is directed to Weidmann's exhaustive text [26].

The Dirac-von Neumann axioms restrict the operators of interest in Quantum Mechanics to those that are self-adjoint and defined on a Hilbert space. In the finite-dimensional case this is equivalent to demanding the matrix representation of the operator corresponding to some observable be Hermitian (i.e. equal to its own conjugate transpose), yielding necessarily real eigenvalues. We recall the definition of an eigenvalue and eigenvector of an operator $\hat{f}$ are a number $\lambda \in \mathbb{C}$ and $v_\lambda \in \mathbb{D}_f$ respectively, with the property that

$$(\hat{f} - \lambda \mathbb{I})v_\lambda = 0. \tag{2.1}$$

As our operator is self-adjoint, its spectrum will necessarily lie on the real line. Often, to ensure stability, we require that the Hamiltonian be bounded from below, which will necessitate that the spectrum have a minimum value, i.e. a ground state. The spectrum can be treated carefully via a decomposition into three parts: a point spectrum σ_p, an absolutely continuous spectrum σ_c, and a singular spectrum σ_s.

A classic result known as the RAGE theorem (due to Ruelle, Amrein-Georgescu, and Enss) uses ergodic methods to relate the long-time behaviour of solutions to quantum mechanical equations to the spectral properties of their Hamiltonians. In particular, RAGE tells us that the point spectrum corresponds to bound states of the system, while the absolutely continuous spectrum corresponds to free or scattering states (the singular spectrum has no easily identifiable physical significance).[1] Moreover, RAGE allows an identification of bound states with time-periodic solutions, which will become important to us when investigating the properties of quantum mechanical wave equations on curved space.

Remark 1. *We can immediately observe some characteristics of the point spectrum. In a separable Hilbert space the point spectrum is necessarily finite or countable because the eigenvectors corresponding to different eigenvalues are orthogonal and any set of non-zero orthogonal vectors in a separable Hilbert space is at most countable.*

In the case where the absolutely continuous part of the spectrum of some operator $\hat{O}$ is empty, denoted $\sigma_c(\hat{O}) = \emptyset$, the operator is said to have pure point spectrum, a

[1] See the almost Mathieu and random Schrödinger operators for examples in which the singular spectrum may have a physical interpretation.

physical example of which is the Schrödinger Hamiltonian for the harmonic oscillator. Likewise when $\sigma_p(\hat{O}) = \emptyset$ the operator is said to have purely absolutely continuous spectrum. The position operator of a free particle moving on a line is the most simple and natural example of a quantum mechanical operator with purely absolutely continuous spectrum; its spectrum is the whole real line. This point will be exploited presently to demonstrate the non-existence of bound states for the massless Dirac-style operator of our photon wave equation.

The position operator $\hat{x}$ is unitarily equivalent to the momentum operator $\hat{p}$. This means that there exists a unitary operator $\hat{F}$ on $\mathcal{L}_2(-\infty, \infty)$ such that

$$\hat{p} = \hat{F}^{-1}\hat{x}\hat{F}. \tag{2.2}$$

The unitary operator $\hat{F}$ that satisfies Eq. 2.2 is the Fourier transform, which should be familiar for the reader who has been exposed to the equivalence of momentum and position-space representations. In more physical language $\hat{p}$ acting on momentum-space wavefunctions is the same as $\hat{x}$ acting on position-space wavefunctions. Moreover, we have:

Proposition 1. *If $\hat{S}$ and $\hat{T}$ are unitarily equivalent operators in a Hilbert space $\mathcal{H}$, then the spectrum of $\hat{S}$ is the same as that of $\hat{T}$.*

Proof. This follows from the definition of unitary equivalence given above. □

So, since we now know that the spectrum of the position operator is purely absolutely continuous, and the momentum operator is unitarily equivalent to the position operator, its spectrum is purely absolutely continuous as well. That is, for $\hat{X} = x$ $\sigma_p(\hat{x}) = \emptyset$ which gives $\sigma_p(\hat{p}) = \emptyset$ for $\hat{p} = -i\hbar\frac{\partial}{\partial x}$.

We now make the transition to an operator that encodes spin, by introducing the Pauli matrices σ_i where the σ_i represents the observable corresponding to spin along the ith coordinate axis.

Proposition 2. *For any potential V that is locally integrable, $-i\sigma_2\frac{d}{dx} + V(x)$ has a purely absolutely continuous spectrum covering all of $\mathbb{R}$.*

Proof. Let U be locally continuous with $U' = V$. Then

$$e^{-i\sigma_2 U(x)}(-i\sigma_2\frac{d}{dx})e^{i\sigma_2 U(x)} = -i\sigma_2\frac{d}{dx} + V(x) \tag{2.3}$$

so V can be removed by a unitary transformation, and hence the full operator with potential has the same spectrum as the momentum operator. □

The operator in question in Proposition 2 is exactly the one-dimensional massless Dirac operator, H_1. This yields the not-so-surprising realization that a charged massless particle in 1D has exactly the same spectrum as an uncharged one, and hence its behaviour will be qualitatively similar. By separation in spherical coordinates this result also implies $\sigma_c(H_D) = \mathbb{R}$ for $D = 2, 3$ if V is rotationally symmetric. This is sufficient to conclude that our photon wave equation has no bound states. This author feels that this is a much stronger and more illuminating result than the argument due to scale invariance presented in Chapter 1.

With an identification of parts of the spectral decomposition with the long-time dynamics of physical systems, as well as a deeper understanding of the absence of bound states in the potential formulation for our photon wave equation, we can now attempt to construct our quantum mechanical wave equation in the context of general relativity. This treatment will *not* be an investigation of the fully coupled Dirac-Einstein equations wherein the back-reaction (a modification of the background due to the particle's energy) is taken into account and the metric is dynamic. In such a setting the energy-momentum tensor in Einstein's equations are sourced by the Dirac particle, and the established metric is then embedded in the definition of the covariant derivative for the Dirac equation. Instead we will treat a Dirac-like particle on a fixed static background, with the knowledge that this is not the complete description of the system and only an approximation for particles of sufficiently small energy compared to that of the central body.

2.2 The Fully Covariant Dirac Equation in the Schwarzschild Background

In this section we treat the Dirac equation, instead of the explicit photon wave equation, for it is well documented and understood in the literature. This equation will differ from the photon wave equation only in the definition of the gamma matrices and the form of these should not affect the existence of bound states. This treatment is motivated by the equivalence between the photon wave equation and the Dirac equation made clear in Chapter 1.

So, following the normal approach, we define the Dirac equation in a general background space-time [19]:

$$[\gamma^a e_a^\mu(\partial_\mu + \Gamma_\mu) + m]\,\Psi = 0, \tag{2.4}$$

where m is the mass of the Dirac field which we will set to zero going forward in order to consider the dynamics of the properly massless (albeit, spin 1/2) photon — and by extension the massless neutrino — field. Here e^μ_a is the inverse of the tetrad e^a_μ defined with respect to the metric $g_{\mu\nu}$,

$$g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu, \tag{2.5}$$

with $\eta_{ab} = \mathrm{diag}(-1, 1, 1, 1)$ being the Minkowski metric. The γ^a are the gamma/Dirac matrices

$$\gamma^0 = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & -i\sigma^i \\ i\sigma^i & 0 \end{pmatrix}, i = 1, 2, 3, \tag{2.6}$$

where σ^i are the Pauli matrices, and the gamma matrices are related to the metric by the anti-commutation relations:

$$\frac{1}{2}\left(\gamma^a\gamma^b + \gamma^b\gamma^a\right) = g^{ab}. \tag{2.7}$$

Γ_μ is the spin connection given by

$$\Gamma_\mu = \frac{1}{8}\left[\gamma^a, \gamma^b\right] e^\nu_a e_{b\nu;\mu}, \tag{2.8}$$

where $e_{b\nu;\mu} = \partial_\mu e_{b\nu} - \Gamma^\alpha_{\mu\nu} e_{b\alpha}$ is the covariant derivative of $e_{b\nu}$ and $\Gamma^\alpha_{\mu\nu}$ are the Christoffel symbols associated with the metric $g_{\mu\nu}$.

2.3 Motivating the Spin Connection

Despite the appearance of all the messy machinery, this really is not much different from the flat-space Dirac equation. All we have done is ensure that our equation is independent of the coordinate system (with our covariant derivative defined using Christoffel symbols) and the spin connection ensures that our spinor Ψ transforms properly on curved space. The necessity of the spin-connection can be motivated if one considers the regular flat-space Dirac equation in the presence of some electromagnetic 4-potential A_μ:

$$i\gamma^\mu\left(\partial_\mu - \frac{iq}{\hbar c}A_\mu\right)\Psi - \frac{mc}{\hbar}\Psi = 0. \tag{2.9}$$

Here the presence of the electromagnetic field mandates the transformation of the simple partial differential operator ∂_μ to $\partial_\mu - \frac{iq}{\hbar c}A_\mu$ which goes by the name of the covariant derivative and is also needed if one is to make the Dirac-Maxwell Lagrangian

gauge invariant. A similar form holds for the spinor in curved space-time, and Weyl realized that this transformation rule holds as a general principle and that we must send $\partial_\mu \to \partial_\mu + \Gamma_\mu$.[2]

2.4 Schwarzschild Coordinates

We now treat our fully covariant Dirac equation on the Schwarzschild space-time introduced in Section 1.4. The line-element is given by:

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2, \tag{2.10}$$

and we can take the tetrad to be

$$e^a_\mu = \text{diag}\left(\left(1 - \frac{2M}{r}\right)^{1/2}, \left(1 - \frac{2M}{r}\right)^{-1/2}, r, r\sin\theta\right), \tag{2.11}$$

where a denotes the basis vector and μ the components of the basis vector with respect to a certain set of coordinate vectors. Then the spin connection can be written as

$$\gamma^a e^\mu_a \Gamma_\mu = \gamma^1 \left(1 - \frac{2M}{r}\right)^{1/2} \left(\frac{1}{r} + \frac{M}{2r(r-2M)}\right) + \gamma^2 \left(\frac{1}{2r}\right) \cot\theta. \tag{2.12}$$

So, now we can write out the explicit form of our Dirac equation on the Schwarzschild background:

$$\begin{aligned}\Bigg[\gamma^0 \left(1 - \frac{2M}{r}\right)^{-1/2} \frac{\partial}{\partial t} + \gamma^1 \left(1 - \frac{2M}{r}\right)^{1/2} \left(\frac{\partial}{\partial r} + \frac{1}{r} + \frac{M}{2r(r-2M)}\right) + \\ \gamma^2 \left(\frac{1}{r}\right) \left(\frac{\partial}{\partial \theta} + \frac{1}{2}\cot\theta\right) + \gamma^3 \left(\frac{1}{r\sin\theta}\right) \frac{\partial}{\partial \phi}\Bigg] \Psi = 0.\end{aligned} \tag{2.13}$$

This is very similar to the Dirac equation in a flat space-time with a central potential. Indeed, the dynamics of the Schwarzschild solution can be captured completely with an effective radial potential, so this is not so surprising. Applying the result from the spectral methods section suggests that due to the spherically symmetric nature of the Schwarzschild space-time, and particularly that of its associated effective potential, we can claim that there are no bound states. This is certainly not rigorous but we motivate this conclusion with another result as follows.

[2] We won't perform a derivation of the field Γ_μ here, but the interested reader is referred to Penrose's 'Spinors and Space-time.'[20]

2.5 Global Rain (Painleve-Gullstrand) Simplification

It turns out that we can choose a coordinate system in which the Dirac equation is particularly simple. The Painleve-Gullstrand coordinates are a frequent choice for describing the Schwarzschild metric as they remove the coordinate singularity (read: event horizon) at $r = 2M$ by the introduction of a new time coordinate $d\tilde{t} = dt + \sqrt{\frac{2M}{r}}\frac{1}{1-\frac{2M}{r}}dr$. The line-element in the Painleve-Gullstrand coordinates is

$$ds^2 = -\left(1 - \frac{2M}{r}\right)d\tilde{t}^2 + 2\sqrt{\frac{2M}{r}}d\tilde{t}dr + dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2 \tag{2.14}$$

and we see that this metric is not diagonal due to the cross term $d\tilde{t}dr$. If we formulate our Dirac equation in these coordinates we are left with an equation of the form

$$i\gamma^\mu \partial_\mu \Psi - i\gamma^0 \sqrt{\frac{2M}{r}}\left(\frac{\partial}{\partial r} + \frac{3}{4r}\right)\Psi = 0. \tag{2.15}$$

This has the advantage of looking exactly like the free-particle Dirac equation plus an interaction term. So, with the choice of coordinates we have converted our problem to Hamiltonian form, although our interaction Hamiltonian H_{int} is non-Hermitian. This result was recognized previously by Lasenby et al. [17], although they leave significant details to the reader. We see that H_{int} is non-Hermitian at the origin due to the presence of the singularity and calculate

$$H_{int} - H_{int}^\dagger = -i\left(2Mr^3\right)^{1/2}\delta(x). \tag{2.16}$$

To achieve the result above, one need only calculate the expectation value of the interaction Hamiltonian and, with integration by parts, it becomes evident that a boundary term dependent on the behaviour of the wave-function must vanish at both $r = 0, +\infty$ in order for the Hamiltonian to be Hermitian. We can easily assume that it vanishes at $+\infty$ in order to demand normalizability. However, the behaviour of the wave-function near the origin is less clear. A messy solution to the differential system using the method of Frobenius shows us that the wave-function behaves like $r^{-3/4}$ near the origin, and hence the boundary term remains, and our Hamiltonian is non-Hermitian. Moreover, we see that this is an acceptable solution in terms of a probabilistic interpretation of the wave-function. As although the wave-function

itself diverges at the origin, the probability density is finite.:

$$P(r)dr \propto r^{-3/4}r^{-3/4}r^2dr = r^{1/2}dr. \tag{2.17}$$

This non-Hermiticity of our Hamiltonian leads us to believe that that the energies associated with any eigenvalues may be complex, and so the state will necessarily *decay* and yield long-term behaviour that looks like infall, thus precluding bound states. It appears this may be due to the singularity acting as a sort-of probability density sink, wherein the complex energies give finite life-times to any eigenstates. This same conclusion was reached via a different and much more rigorous demonstration of the asymptotic behaviour of solutions to the massless-Dirac equation in the Schwarzschild metric carried out by Smoller and Xie in 2011 [24]. In this paper they show an explicit form for the decay rate as $t^{-2\lambda}$ where λ is the angular momentum of the Dirac particle in units of $\hbar$.

We have shown that bound states fail to exist in the spherically symmetric Schwarzschild background for our massless-Dirac equation. For the reasons explained above, this is not particularly surprising, and it seems that none of our hard work in setting up the general relativistic machinery has paid off. However, perhaps we can use this opportunity to probe the role of the rest-mass for the particle in the lack of bound states. In analogy to Dirac hydrogen, where re-insertion of a mass into the operator saves the bound states, it is reasonable to believe that a similar procedure could be performed in the general relativistic setting. However, on closer inspection one notices something a bit odd. The argument carried out directly above for the non-Hermitian form of the Dirac equation in the Painleve-Gullstrand coordinates works *exactly* the same for a massive Dirac particle. That is, we have a small reason to suspect that perhaps even massive particles lack bound states in certain backgrounds. We can, perhaps, motivate this result as follows: In general relativity the geodesics of particles on a background are independent of their mass. Moreover, in the minimal coupling setting in which we have formed the covariant Dirac equation all dependence on mass vanishes. This is a direct consequence of the equivalence principle, and therefore provides a way of explaining the ineffectiveness of re-inserting a mass term in altering the dynamics of the system.

However, while we may be able to explain the equivalence of massive and massless treatments, the absence of bound states is still surprising as the Dirac equation describes fermions, the constituents of all matter, for which we certainly expect gravitationally bound states in the classical setting. In the next chapter we investigate

the properties of the massive Dirac equation in the most general background solution to Einstein's equation: the Kerr-Newman metric, set up by a charged and rotating massive body, in an attempt to better understand the issues at hand.

Chapter 3

A Massive Dirac Particle in the Kerr-Newman Geometry

The Kerr-Newman background is the most general known electrovacuum solution to the Einstein field equation. The line element in the Boyer-Lindquist coordinates is:

$$ds^2 = -\frac{\Delta}{\rho^2}\left(dt - a\sin^2\theta d\phi\right)^2 + \frac{\sin^2\theta}{\rho^2}\left(\left(r^2+a^2\right)d\phi - adt\right)^2 + \frac{\rho^2}{\Delta}dr^2 + \rho^2 d\theta^2, \quad (3.1)$$

with $\rho^2 = r^2 + a^2$, $a = \frac{J}{M}$, and $\Delta = r^2 - 2Mr + a^2 + Q^2$. We build our length-scale (Q) corresponding to the electric charge as $Q^2 = \frac{Gq^2}{4\pi\epsilon_0 c^4}$. This metric describes the space-time surrounding an electrically-charged (q), rotating (angular momentum J) and massive (M) body. We will attempt to investigate the dynamics of a Dirac particle in this background, with the hope that its general nature will allow for a complete picture of the dynamics of quantum wavepackets in the general relativistic setting. That is, this metric contains the Kerr geometry (for $Q = 0$), Reissner-Nordstrom (for $a = 0$), and the Schwarzschild case ($Q = a = 0$). We will show that there exist no non-trivial time-periodic solutions to the Dirac equation in this space-time and that the only long-term behaviours allowed are infall and scattering to spatial infinity. This result seems surprising and raises broad questions as to what our expectations for predictions in this setting should be.

This result was first achieved by Finster et al. [13], and a brief qualitative outline of their approach is presented in Section 3.2. However, our approach will be demonstrated through somewhat different means, first utilized in this setting by Winklmeier and Yamada [28], that we believe to be substantially clearer and more concise.

3.1 Chandrasekhar Separation of the Dirac Equation on Kerr-type Metrics

When attempting to analyse the properties of fields, or solutions to wave equations, in the more realistic framework of Kerr-type metrics, we are faced with an issue that is not present in the simplified picture of the Schwarzschild background. Due to the angular momentum of the massive body, and associated frame dragging, all Kerr-type metrics lose the spherical symmetry that the Schwarzschild solution has and are left with axial symmetry since they have only two commuting Killing vector fields, for the time-like and longitudinal coordinates. However, a stunning result due to Chandrasekhar [8] shows that the Dirac equation *is* separable in the radial and angular coordinates. Using this result one can separate the Dirac partial differential equation into two, uncoupled, linear first order differential equations for the radial and angular parts, with the addition of a separation constant λ. This tool is indispensable as it allows for an isolation of the radial equation necessary for the study of the bound states and time-periodic solutions of the Dirac equation. However, the fact that this separation is possible motivates a question regarding the underlying symmetry behind the separation, as we have stated that this separation is not due to an obvious symmetry of the metric. Indeed, it can be shown that λ is an eigenvalue of a symmetry operator, with the interpretation of the total angular momentum, and is directly related to Carter's constant which is a conserved quantity related to a higher-order symmetry in the Kerr metric. While we will not explicitly show the separation result here, we will use it in the following sections, and hope that the reader finds it plausible, at least, with the brief explanation and motivation given above.[1]

3.2 Method of Finster et al.

Finster et al. derive the Chandrasekhar separation result for Kerr-Newman in the Boyer-Lindquist coordinates, which contains Cauchy and event horizons.[2] In order to properly treat the Dirac equation across these horizons, they transform to Kerr coordinates where the metric, and hence the Dirac operator, becomes regular on the horizons, and derive matching conditions. They then transform these conditions back to Boyer-Linquist, and define the time-periodicity of a wave function in the normal

[1] For an explicit and careful treatment, see Chandrasekhar's monograph 'The Mathematical Theory of Black Holes.'[8]

[2] The Cauchy horizon is the inner horizon of the metric, below which one cannot predict the dynamics of the region without specifying boundary conditions at the singularity.

manner. Finster et al. then proceed to use the particulars of the matching conditions combined with the positivity and conservation of the Dirac current to demonstrate that the only way in which a time-periodic solution can be normalizable is for each term in the Fourier expansion of the wave-function, in time and the angular variable, to be identically zero.

3.3 Forming the Massive Dirac Equation on the Kerr-Newman Background

For the Dirac equation describing a spin-$\frac{1}{2}$ particle with mass m, intrinsic angular moment J and charge e in the Kerr-Newman metric we have [13]

$$(\hat{R} + \hat{A})\Psi = 0, \tag{3.2}$$

where

$$\hat{R} := \begin{bmatrix} imr & 0 & \sqrt{\Delta}D_+ & 0 \\ 0 & -imr & 0 & \sqrt{\Delta}D_- \\ \sqrt{\Delta}D_- & 0 & -imr & 0 \\ 0 & \sqrt{\Delta}D_+ & 0 & imr \end{bmatrix} \tag{3.3}$$

$$\hat{A} := \begin{bmatrix} -am\cos\theta & 0 & 0 & \mathcal{L}_+ \\ 0 & am\cos\theta & -\mathcal{L}_- & 0 \\ 0 & \mathcal{L}_+ & -am\cos\theta & 0 \\ -\mathcal{L}_- & 0 & 0 & am\cos\theta \end{bmatrix} \tag{3.4}$$

$$\mathcal{D}_\pm := \frac{\partial}{\partial r} \mp \frac{1}{\Delta}\big[(r^2 + a^2)\frac{\partial}{\partial t} + a\frac{\partial}{\partial \phi} - ieQr\big] \tag{3.5}$$

$$\mathcal{L}_\pm := \frac{\partial}{\partial \theta} + \frac{\cot\theta}{2} \mp i\big[a\sin\theta\frac{\partial}{\partial t} + \frac{1}{\sin\theta}\frac{\partial}{\partial \phi}\big] \tag{3.6}$$

$$\Delta(r) := r^2 - 2Mr + a^2 + Q^2 \tag{3.7}$$

here Ψ is the spinor for our spin-$\frac{1}{2}$ particle. In order to make predictions about future evolution around the black hole we must disallow naked singularities[3] — singularities not clothed in an event horizon — and so we constrain our three parameters as follows:

$$M^2 - a^2 - Q^2 \geq 0, \tag{3.8}$$

[3]Else we may end up with 'green slime and lost socks' a la Earman. Really this has to do with the lack of uniqueness of solutions to the boundary value problem in the presence of a naked singularity.

which allows Δ to be written as

$$\Delta(r) = (r - r_-)(r - r_+) \tag{3.9}$$

with

$$r_\pm = M \pm \sqrt{M^2 - a^2 - Q^2}. \tag{3.10}$$

These two radii correspond to the event horizon of the metric, and the internal Cauchy horizon. For the extremal case where $M^2 - a^2 - Q^2 = 0$ these horizons become degenerate with $r_+ = M = r_-$. In the demonstration that follows we will focus on the non-extremal case. The existence of time-periodic normalizable solutions under certain conditions has been shown in the extremal case by Schmid [21].

3.4 Separation of the Dirac Equation

We now focus on time-periodic solutions and we define our periodicity conditions as follows

$$\Psi(r, \theta, \phi, t) = \Psi(r, \theta, \phi, t + \frac{2\pi}{\omega_0}) \tag{3.11}$$

$$\Psi(r, \theta, \pi, t) = -\Psi(r, \theta, -\pi, t) \tag{3.12}$$

for some $\omega_0 > 0$. We proceed to show that the only solution to the Dirac equation satisfying these conditions in the non-extremal Kerr-Newman metric is the solution $\Psi \equiv 0$. We begin by expanding Ψ as a Fourier series in t

$$\Psi(r, \theta, \phi, t) = \sum_{\nu \in \mathcal{Z}} e^{-i\omega_0 \nu t} \Psi^\nu(r, \theta, \phi) \tag{3.13}$$

where the Ψ^ν are defined as

$$\Psi^\nu(r, \theta, \phi) = \frac{\omega_0}{2\pi} \int_0^{2\pi/\omega_0} e^{i\omega_0 \nu t} \Psi(r, \theta, \phi, t) dt. \tag{3.14}$$

Demanding physicality means that our wave functions Ψ^ν must be elements of the Hilbert space spanning $-\pi \to \pi$ in the ϕ coordinate, $0 \to \pi$ in the θ coordinate and $r_+ \to \infty$ in the radial coordinate, where Finster et al. [13] define the inner product as

$$\langle \Psi, \Phi \rangle = \int_{r_+}^{\infty} \int_0^{\pi} \int_{-\pi}^{\pi} \langle \Psi(r, \theta, \phi), \Phi(r, \theta, \phi) \rangle_{\mathbb{C}^4} \frac{r^2 + a^2}{\Delta(r)} \sin\theta dr d\theta d\phi, \tag{3.15}$$

with $\langle \Psi, \Phi \rangle_{\mathbb{C}^4}$ the normal scalar product on $\mathbb{C}^4$. We can further expand the Ψ^ν as

$$\Psi^\nu(r,\theta,\phi) = \sum_{\kappa \in \mathcal{Z}+\frac{1}{2}} e^{-i\kappa\phi} \Psi^{\nu,\kappa}(r,\theta), \tag{3.16}$$

where the κ index just keeps track of our separation of the ϕ dependence, which is justified given the axial symmetry of the geometry. So, if we have a

$$\Psi(r,\theta,\phi,t) = \sum_{\nu \in \mathbb{Z}} \sum_{\kappa \in \mathbb{Z}+\frac{1}{2}} e^{-i\omega_0 \nu t} e^{-i\kappa\phi} \Psi^{\nu,\kappa}(r,\theta) \tag{3.17}$$

that satisfies the general Dirac equation in Eq. 3.2, then we can say that each $\Psi^{\nu,\kappa}$ is itself a solution to

$$(\mathcal{R}_{\nu,\kappa} + \mathcal{A}_{\nu,\kappa})\Psi^{\nu,\kappa} = 0 \tag{3.18}$$

where $\mathcal{R}_{\nu,\kappa}$ and $A_{\nu,\kappa}$ are given in the same form as in Eq. 3.3 and Eq. 3.4 with the following replacement of the differential operators $\mathcal{D}_\pm$ and $\mathcal{L}_\pm$:

$$\mathcal{D}_{\pm,\nu,\kappa} := \frac{\partial}{\partial r} \pm \frac{i}{\Delta}\big(\omega_0 \nu (r^2 + a^2) + \kappa a + eQr\big) \tag{3.19}$$

$$\mathcal{L}_{\pm,\nu,\kappa} := \frac{\partial}{\partial \theta} + \frac{\cot\theta}{2} \mp \big(a\omega_0 \nu \sin\theta + \frac{\kappa}{\sin\theta}\big). \tag{3.20}$$

It has been shown that the operator $A_{\nu,\kappa}$ has a purely discrete spectrum $\sigma(A_{\nu,\kappa}) = (\lambda : n \in \mathbb{Z} \setminus 0)$ where the eigenvalues correspond to discrete states of angular momentum [3]. So, there is a complete set of orthonormal eigenfunctions of $A_{\nu,\kappa}$, with associated eigenvalues $\lambda_{\nu,\kappa,n}$, given by

$$z^{\nu,\kappa,n}(\theta) = \begin{pmatrix} z_1^{\nu,\kappa,n}(\theta) \\ z_2^{\nu,\kappa,n}(\theta) \end{pmatrix}. \qquad (n \in \mathbb{Z} \setminus 0) \tag{3.21}$$

This allows us to make the ansatz

$$\Psi^{\nu,\kappa} = \sum_{n \in \mathbb{Z}\setminus 0} \Psi^{\nu,\kappa,n} \tag{3.22}$$

where

$$\Psi^{\nu,\kappa,n}(r,\theta) = \begin{pmatrix} X_-^{\nu,\kappa,n} z_2^{\nu,\kappa,n}(\theta) \\ X_+^{\nu,\kappa,n} z_1^{\nu,\kappa,n}(\theta) \\ X_+^{\nu,\kappa,n} z_2^{\nu,\kappa,n}(\theta) \\ X_-^{\nu,\kappa,n} z_1^{\nu,\kappa,n}(\theta) \end{pmatrix} \qquad (r \in (r_+,\infty), \theta \in (0,\pi)) \tag{3.23}$$

which, via the method of Chandrasekhar, leads to a separation of the radial and angular coordinates. The angular function $z^{\nu,\kappa,n}$ satisfies the simple angular eigenvalue equation

$$A_{\nu,n} z^{\nu,\kappa,n} = \lambda_{\nu,\kappa,n} z^{\nu,\kappa,n}, \tag{3.24}$$

and the radial function $X^{\nu,\kappa,n}$ satisfies the more complex radial equation

$$\begin{pmatrix} -imr - \lambda_{\nu,\kappa,n} & \sqrt{\Delta}\mathcal{D}_{-,\nu,\kappa} \\ \sqrt{\Delta}\mathcal{D}_{+,\nu,\kappa} & imr - \lambda_{\nu,\kappa,n} \end{pmatrix} \begin{pmatrix} X_+^{\nu,\kappa,n}(r) \\ X_-^{\nu,\kappa,n}(r) \end{pmatrix} = 0. \tag{3.25}$$

3.5 The Radial Equation

In this section we will perform a coordinate transformation on the radial Hamiltonian separated in the preceding section, so as to put it into a form whereby we can use existing spectral results. Let

$$K := \frac{1}{\sqrt{2}} \begin{pmatrix} -i & i \\ -1 & -1 \end{pmatrix}, \qquad J := \begin{pmatrix} 0 & -1/\sqrt{\Delta} \\ 1/\sqrt{\Delta} & 0 \end{pmatrix}, \qquad \tilde{f}^{\nu,\kappa,n} := K \begin{pmatrix} X_+^{\nu,\kappa,n}(r) \\ X_-^{\nu,\kappa,n}(r) \end{pmatrix}. \tag{3.26}$$

Then, from Eq. 3.25 we obtain

$$0 = JK \begin{pmatrix} -imr - \lambda_{\nu,\kappa,n} & \sqrt{\Delta}\mathcal{D}_{-,\nu,\kappa} \\ \sqrt{\Delta}\mathcal{D}_{+,\nu,\kappa} & imr - \lambda_{\nu,\kappa,n} \end{pmatrix} \begin{pmatrix} X_+^{\nu,\kappa,n}(r) \\ X_-^{\nu,\kappa,n}(r) \end{pmatrix} K^{-1} \tilde{f}^{\nu,\kappa,n}(r) \tag{3.27}$$

$$= \begin{pmatrix} \frac{mr}{\sqrt{\Delta}} - \frac{a\kappa + eQr + \omega_0\nu(r^2+a^2)}{\Delta} & \frac{-d}{dr} + \frac{\lambda_{\nu,\kappa,n}}{\sqrt{\Delta}} \\ \frac{d}{dr} + \frac{\lambda_{\nu,\kappa,n}}{\sqrt{\Delta}} & -\frac{mr}{\sqrt{\Delta}} - \frac{a\kappa + eQr + \omega_0\nu(r^2+a^2)}{\Delta} \end{pmatrix} \tilde{f}^{\nu,\kappa,n}(r) \tag{3.28}$$

for $r \in (r_+,\infty)$. It is useful to note here that if we take terms that are only of first order in $1/r$ for large r, then Eq. 3.27 reduces exactly to the radial equation for Dirac-Hydrogen:

$$\begin{pmatrix} m - \frac{eQ}{r} + \omega_0\nu & -\frac{d}{dr} + \frac{\lambda_{\nu,\kappa,n}}{r} \\ \frac{d}{dr} + \frac{\lambda_{\nu,\kappa,n}}{r} & -m - \frac{eQ}{r} + \omega_0\nu \end{pmatrix} \tilde{f}(r) = 0. \tag{3.29}$$

In order to put our radial equation in a form such that we can utilise spectral results in the literature we introduce a coordinate x such that

$$\frac{dx}{dr} = \frac{r^2 + a^2}{\Delta(r)} \qquad (r > r_+). \tag{3.30}$$

Explicitly, this is

$$x(r) = r + \frac{r_+^2 + a^2}{r_+ - r_-}\ln(r - r_+) - \frac{r_+^2 + a^2}{r_+ - r_-}\ln(r - r_-) + x_0 \qquad r_+ \neq r_- \tag{3.31}$$

where we can set $x_0 = 0$ as our constant of integration. In these new coordinates Eq. 3.27 becomes

$$H_{\nu,\kappa,n} f^{\nu,\kappa,n} := \begin{pmatrix} \frac{mr\sqrt{\Delta}}{r^2+a^2} - \frac{a\kappa+eQr}{r^2+a^2} & -\frac{d}{dx} + \frac{\lambda_{\nu,\kappa,n}\sqrt{\Delta}}{r^2+a^2} \\ \frac{d}{dx} + \frac{\lambda_{\nu,\kappa,n}\sqrt{\Delta}}{r^2+a^2} & -\frac{mr\sqrt{\Delta}}{r^2+a^2} - \frac{a\kappa+eQr}{r^2+a^2} \end{pmatrix} f^{\nu,\kappa,n} = \omega_0 \nu f^{\nu,\kappa,n}. \tag{3.32}$$

3.6 Spectral Analysis on our Hamiltonian of Interest

Our eigenvalue equation of interest is $(H_{\nu,\kappa,n} - \omega_0\nu)f^{\nu,\kappa,n} = 0$ from Eq. 3.32. In this section we will show that our Hamiltonian operator has a purely absolutely continuous spectrum, except at one point, and then that this point is still *not* an eigenvalue of our operator. As such, the point spectrum of our Hamiltonian is empty and we have no bound states. We begin by decomposing our operator H into a sum of operators in order to exploit existing theorems:

$$H = H_0 + V$$

with

$$H_0 = \begin{pmatrix} 0 & -\frac{d}{dx} \\ \frac{d}{dx} & 0 \end{pmatrix}, \qquad V(x) = \begin{pmatrix} A(x) & B(x) \\ B(x) & C(x) \end{pmatrix}$$

$$\begin{aligned} A(x) &= \frac{mr(x)\sqrt{\Delta(r(x))}}{r(x)^2+a^2} - \frac{a\kappa + eQr(x)}{r(x)^2+a^2} \\ B(x) &= \lambda \frac{\Delta(r(x))}{r(x)^2+a^2} \\ C(x) &= -\frac{mr(x)\sqrt{\Delta(r(x))}}{r(x)^2+a^2} - \frac{a\kappa + eQr(x)}{r(x)^2+a^2} \\ \Delta(x) &= (r(x)-r_+)(r(x)-r_-). \end{aligned}$$

H_0 is just the massless free-space Dirac operator, whose spectrum we are familiar with, and it is symmetric and self-adjoint. We now show that the operator V has certain properties such that it does not endow the entire operator with eigenvalues. We can easily see that A, B, C are bounded. With $x \to -\infty$ equivalent to $r(x) \to r_+$ and $x \to \infty$ equivalent to $r(x) \to \infty$, we have:

$$\lim_{x\to-\infty} A(x) = -\frac{a\kappa + eQr_+}{r_+^2+a^2} = \lim_{x\to-\infty} C(x) := A_0$$

$$\lim_{x\to+\infty} A(x) = m \qquad \lim_{x\to+\infty} C(x) = -m \qquad \lim_{x\to\pm\infty} B(x) = 0,$$

where A_0 is just a constant. V being a symmetric and bounded perturbation of H_0 allows us to conclude that H is also self-adjoint. In the work that follows we will assume the non-extremal case $r_+ \neq r_-$.

Lemma 1. *Asymptotic behaviour of V(x).*
(a) As $x \to -\infty$ the functions $A(x) - A_0$, $B(x)$, $C(x) - A_0$ decay exponentially.
(b) A', B', C' are integrable in x on the interval $(-\infty, 0]$.
(c) As $x \to \infty$, $A' = B' = C' = O(x^{-2})$.

Proof. From our definition of the new coordinate x in Eq. 3.31 we see that

$$r(x) - r_+ = O\left(e^{\alpha x}\right) \text{ as } x \to -\infty. \tag{3.33}$$

(a) If one notes that

$$\frac{1}{r^2+a^2} = \frac{1}{r_+^2+a^2} - \frac{(r-r_+)(r+r_+)}{(r_+^2+a^2)(r^2+a^2)} = \frac{1}{r_+^2+a^2} + O(r-r_+), \tag{3.34}$$

then we see immediately that

$$B(x) = \lambda \frac{\sqrt{\Delta(r)}}{r^2 + a^2} = \lambda \sqrt{\Delta(r)} \left(\frac{1}{r_+^2 + a^2} + O(r - r_+) \right). \tag{3.35}$$

Then using the result from Eq. 3.34 and the definition of $\Delta(r)$ we see that the assertion for B is evident. To prove the result for A we show

$$\begin{aligned} A(x) - A_0 &= \frac{mr\sqrt{\Delta(r)}}{r^2 + a^2} + \frac{(r^2 + a^2)(a\kappa + eQr_+) - (r_+^2 + a^2)(a\kappa + eQr)}{(r^2 + a^2)(r_+^2 + a^2)} \\ &= \frac{mr\sqrt{\Delta(r)}}{r^2 + a^2} + \frac{(r - r_+)}{(r^2 + a^2)} \frac{(r - r_+)(a\kappa + eQr_+) + 2a\kappa r_+ + eQ(r_+^2 - a^2)}{(r_+^2 + a^2)} \\ &= \frac{mr_+\sqrt{\Delta(r)}}{r^2 + a^2} + \frac{r - r_+}{r^2 + a^2} \left[m\sqrt{\Delta(r)} + \frac{(r - r_+)(a\kappa + eQr_+) + 2a\kappa r_+ + eQ(r_+^2 - a^2)}{r_+^2 + a^2} \right], \end{aligned} \tag{3.36}$$

where one can use Eq. 3.33 and Eq. 3.34 to easily demonstrate the truth of our assertion for A. The same assertion can be easily demonstrated for C by replacing $m \to -m$ in the argument above.

(b) Calculemus!

$$\begin{aligned} \frac{d}{dx} A(x) &= \frac{dr}{dx} \frac{d}{dr} A(x(r)) = \frac{\Delta(r)}{r^2 + a^2} \frac{d}{dr} A(x(r)) \\ &= \frac{\sqrt{\Delta(r)}}{r^2 + a^2} \left[m \left(\frac{\Delta(r)}{r^2 + a^2} + \frac{r\Delta'(r)}{2(r^2 + a^2)} - \frac{2r^2\Delta(r)}{(r^2 + a^2)^2} \right) + \frac{2(a\kappa + eQ)r\sqrt{\Delta(r)}}{(r^2 + a^2)^2} - \frac{eQ\sqrt{\Delta(r)}}{r^2 + a^2} \right] \\ &= O((r - r_+)^{1/2}) = O\left(e^{\frac{1}{2}\alpha x}\right) \end{aligned}$$

as $x \to -\infty$. Likewise

$$\begin{aligned} \frac{d}{dx} B(x) &= \frac{dr}{dx} \frac{d}{dr} B(x(r)) = \frac{\Delta(r)}{r^2 + a^2} \frac{d}{dr} B(x(r)) \\ &= \lambda \frac{\sqrt{\Delta(r)}}{r^2 + a^2} \left[\frac{\Delta'(r)}{2(r^2 + a^2} - \frac{2r\Delta(r)}{(r^2 + a^2)^2} \right. \\ &= O((r - r_+)^{1/2}) = O\left(e^{\frac{1}{2}\alpha x}\right) \end{aligned}$$

as $x \to -\infty$.

(c) As

$$\frac{dr}{dx} = \frac{\Delta(r)}{r^2 + a^2} \sim 1 \text{ and } x \sim r \text{ as } x \to +\infty$$

and

$$2(r^2+a^2)\Delta(r)+r(r^2+a^2)\Delta^{'}(r)-4r^2\Delta(r) = 2Mr^3+2r^3(a^2-Q^2)-6a^2Mr+2a^2(a^2+Q^2) \sim r^3$$

as $r \to \infty$ the assertions follow clearly from the proof of part (b) of our lemma. □

We remark that it has previously been demonstrated that the essential spectrum of H is the Real line [4]. This proof relies necessarily on the fact that

$$\lim_{x\to-\infty} [V(x) - A_0 I_2] = 0. \tag{3.37}$$

Theorem 1. *(a) H has a purely absolutely continuous spectrum in $\mathbb{R} \setminus A_0$.*

Proof. Using Lemma 1 on the asymptotic behaviour of V we can get Eq. 3.37, and the fact that all components of V' are integrable at $-\infty$ we can use a result from Thaller [25](Theorem 4.18) to prove this assertion. □

Theorem 2. *A_0 is not an eigenvalue of H.*

Proof. Let there be a $P = (p_1, p_2)$ that satisfies

$$\begin{pmatrix} A & B - \frac{d}{dx} \\ B + \frac{d}{dx} & C \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = A_0 \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} \tag{3.38}$$

so

$$P' = \begin{pmatrix} p_1^{'} \\ p_2^{'} \end{pmatrix} = \begin{pmatrix} -B & A_0 - C \\ A - A_0 & B \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}. \tag{3.39}$$

We have demonstrated in our earlier lemma that $A(x) - A_0$, $C(x) - A_0$, $B(x)$ all decay exponentially as $x \to -\infty$. So, by the Levinson theorem [11] there exist two linearly independent solutions P^1, P^2 to the above equation such that

$$P^1(x) = \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + O(1) \right] \text{ as } x \to -\infty. \tag{3.40}$$

$$P^2(x) = \left[\begin{pmatrix} 0 \\ 1 \end{pmatrix} + O(1) \right] \text{ as } x \to -\infty. \tag{3.41}$$

So there exist constants c_1 and c_2 such that $P(x) = c_1 P^1(x) + c_2 P^2(x)$, which is only square integrable on $(-\infty, 0)$ for $c_1 = c_2 = 0$. □

3.7 Non-Existence of Time-Periodic Solutions

In the preceding sections we separated the radial Hamiltonian for our massive Dirac particle, and then showed that this Hamiltonian has empty point spectrum. In this section we arrive at the same Hamiltonian via a different approach in which we impose our time-periodic conditions: For all Φ of the form

$$\Phi(r,\theta) = \begin{pmatrix} \rho_-(r) z_2(\theta) \\ \rho_+(r) z_1(\theta) \\ \rho_+(r) z_2(\theta) \\ \rho_-(r) z_1(\theta) \end{pmatrix} \tag{3.42}$$

we can obtain, using the periodicity conditions in Eq. 3.11, the following equality.

$$\begin{aligned} 0 &= \int_0^{2\pi/\omega_0} dt \int_{-\pi}^{\pi} d\phi \big\langle (\hat{\mathcal{R}} + \hat{\mathcal{A}})\Psi, exp(-i\omega_0 \nu t) exp(-i\kappa\phi)\Phi \big\rangle \\ &= \int_0^{2\pi/\omega_0} dt \int_{-\pi}^{\pi} d\phi \big\langle \Psi, (\hat{\mathcal{R}} + \hat{\mathcal{A}})^* exp(-i\omega_0 \nu t) exp(-i\kappa\phi)\Phi \big\rangle \\ &= \int_0^{2\pi/\omega_0} dt \int_{-\pi}^{\pi} d\phi \big\langle \Psi, exp(-i\omega_0 \nu t) exp(-i\kappa\phi)(\mathcal{R}_{\nu,\kappa} + \mathcal{A}_{\nu,\kappa})^* \Phi \big\rangle \\ &= \int_0^{2\pi/\omega_0} \big\langle \Psi, exp(-i\omega_0 \nu t) exp(-i\kappa\phi)(\mathcal{R}_{\nu,\kappa} + \mathcal{A}_{\nu,\kappa})^* \Phi \big\rangle dt \\ &= \frac{2\pi}{\omega_0} \int_{-\pi}^{\pi} \big\langle \Psi^\nu, exp(-i\kappa\phi)(\mathcal{R}^*_{\nu,\kappa} + \mathcal{A}^*_{\nu,\kappa})\Phi \big\rangle d\phi \\ &= \frac{4\pi^2}{\omega_0} \big\langle \Psi^{\nu,\kappa}, (\mathcal{R}^*_{\nu,\kappa} + \mathcal{A}^*_{\nu,\kappa})\Phi \big\rangle. \end{aligned} \tag{3.43}$$

Again we take our eigenfunctions of $A_{\nu,\kappa}$, the $z(\theta)$, which allow $\Psi^{\nu,\kappa}(r,\theta)$ to be expanded in terms of our $z^{\nu,\kappa,n}$ as

$$\Psi^{\nu,\kappa}(r,\theta) = \sum_{n \in \mathbb{Z}\setminus 0} \begin{pmatrix} X_-^{\nu,\kappa,n}(r) z_2^{\nu,\kappa,n}(\theta) \\ X_+^{\nu,\kappa,n}(r) z_1^{\nu,\kappa,n}(\theta) \\ X_+^{\nu,\kappa,n}(r) z_2^{\nu,\kappa,n}(\theta) \\ X_-^{\nu,\kappa,n}(r) z_1^{\nu,\kappa,n}(\theta) \end{pmatrix}. \tag{3.44}$$

Substituting our expression for the form of Φ in Eq. 3.42 into our result from Eq.

3.43 we obtain

$$\left\langle \sum_{n\in\mathbb{Z}\setminus 0} \begin{pmatrix} X_-^{\nu,\kappa,n}(r) z_2^{\nu,\kappa,n}(\theta) \\ X_+^{\nu,\kappa,n}(r) z_1^{\nu,\kappa,n}(\theta) \\ X_+^{\nu,\kappa,n}(r) z_2^{\nu,\kappa,n}(\theta) \\ X_-^{\nu,\kappa,n}(r) z_1^{\nu,\kappa,n}(\theta) \end{pmatrix}, (R^*_{\nu,\kappa} - \lambda_{\nu,\kappa\, n}) \begin{pmatrix} \rho_-(r) z_2^{\nu,\kappa,n}(\theta) \\ \rho_+(r) z_1^{\nu,\kappa,n}(\theta) \\ \rho_+(r) z_2^{\nu,\kappa,n}(\theta) \\ \rho_-(r) z_1^{\nu,\kappa,n}(\theta) \end{pmatrix} \right\rangle = 0. \tag{3.45}$$

With the definition of the inner product in Eq. 3.15 we get

$$\int_{r_+}^{\infty} dr \frac{r^2+a^2}{\Delta(r)} \left\langle \begin{pmatrix} X_+^{\nu,\kappa,n} \\ X_-^{\nu,\kappa,n} \end{pmatrix}, \begin{pmatrix} imr - \lambda_{\nu,\kappa,n} & \mathcal{D}^*_{+,\nu,\kappa}\sqrt{\Delta} \\ \mathcal{D}^*_{-,\nu,\kappa}\sqrt{\Delta} & -imr - \lambda_{\nu,\kappa,n} \end{pmatrix} \begin{pmatrix} \rho_+ \\ \rho_- \end{pmatrix} \right\rangle = 0. \tag{3.46}$$

We move our operator over, taking its conjugate transpose in the process, and find the condition

$$\begin{pmatrix} -imr - \lambda_{\nu,\kappa,n} & \sqrt{\Delta}\mathcal{D}_{-,\nu,\kappa} \\ \sqrt{\Delta}\mathcal{D}_{+,\nu,\kappa} & imr - \lambda_{\nu,\kappa,n} \end{pmatrix} \begin{pmatrix} X_+^{\nu,\kappa,n}(r) \\ X_-^{\nu,\kappa,n}(r) \end{pmatrix} = 0. \tag{3.47}$$

We can now set

$$f^{\nu,\kappa,n}(x) = \begin{pmatrix} f_1^{\nu,\kappa,n}(r(x)) \\ f_2^{\nu,\kappa,n}(r(x)) \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -i & i \\ -1 & -1 \end{pmatrix} \begin{pmatrix} X_+^{\nu,\kappa,n}(r(x)) \\ X_-^{\nu,\kappa,n}(r(x)) \end{pmatrix} \tag{3.48}$$

to get

$$H_{\nu,\kappa,n} f^{\nu,\kappa,n} = \omega_0 \nu f^{\nu,\kappa,n}. \tag{3.49}$$

We showed in Section 3.6 that this Hamiltonian has empty point spectrum and hence no bound states.

3.8 Wrapping Up

For those that got lost in the technical details, we will give a short qualitative description of the procedure carried out to demonstrate the absence of time-periodic solutions and bound states. We first set up the form of our Dirac equation on the Kerr-Newman background and separate out the temporal and ϕ dependence in our wavefunction. We then exploit the Chandrasekhar separation result to write our wavefunction as the product of radial functions, the $X_{+,-}$'s and functions in θ, the $z_{2,1}$'s. From this we find a form for our radial equation, whose Hamiltonian we massage into a form amenable for analysis with a coordinate transformation. We proceed to show that

this Hamiltonian can be decomposed into pieces with known spectral properties, and we show that the spectrum of our radial Hamiltonian is purely absolutely continuous; that is, its point spectrum is empty. Then, we see that an imposition of our time-periodic conditions as well as the definition of the inner product on the space yield the same Hamiltonian with empty point spectrum. So, we have a doubly rigorous demonstration of our result. On one hand we show that the radial equation has no bound states, which we would have deemed acceptable as the basis for a claim that there are no time-periodic solutions via the RAGE theorem. Separately we show that imposing time-periodic solutions leads to the same empty point spectrum. The physical implications of this result is that in the limit $t \to \infty$ there is no matter present around a black hole, it must all experience infall past the event horizon or escape to infinity.

So, we have come a long way in our story. We started with an attempt to save the classical orbit of light around a massive body in the Newtonian setting, and have just arrived at a demonstration that there exist no bound states for a massive Dirac particle in the most general of backgrounds (one which corresponds to the metric set up by a hydrogen atom, if more shock value is needed!). Following the general progression of this thesis it would fit now to treat the fully coupled Einstein-Dirac system and attempt to save our bound states in this new setting. However, this type of treatment is outside the range of this author's abilities and so, in the next chapter, we will instead attempt to probe the issue within the fixed-background setting at hand.

Chapter 4

Probing the Bound State Problem in the General Relativistic Setting

Our problem to consider is as follows: In the general relativistic setting there are certainly time-periodic solutions, why does the introduction of a quantized Hamiltonian disallow these? There are a few obvious possible culprits for investigation. Perhaps the introduction of the intrinsic spin angular momentum and an associated coupling to the metric prevents bound states. In addition, the aforementioned result regarding the *existence* of bound states in the extreme Kerr-Newman metric (where, we remind the reader, the two horizons become degenerate) suggests that perhaps the horizons are part of the issue. We probe these questions in this chapter, hoping to come away with a better understanding of the result obtained in Chapter 2, and of the necessary conditions for the existence of a bound state for the Dirac equation in the general relativistic setting.

4.1 Bound States for a Spin-less Scalar Field

Throughout this thesis we have treated relativistic quantum mechanical wave equations for particles with intrinsic spin angular momentum. However, there exists a manifestly Lorentz covariant wave equation suitable for spin-less, or scalar, fields. This equation is known as the Klein-Gordon equation, and while its applications are limited due to the lack of elementary particles with spin-0 (save for the Higgs Boson), we will utilize it to probe what role, if any, spin plays in killing our bound states. The Klein-Gordon equation is of second-order in both space and time, which follows naturally from its derivation from the specially relativistic energy-momentum relation, and so can be loosely thought of as the square of the Dirac operator.

The generally covariant Klein-Gordon equation for a massive scalar field Φ of spin-0 on a background with metric $g_{\mu\nu}$ is given by:

$$\frac{1}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}g^{\mu\nu}\partial_\nu\Phi\right)+\mu^2\Phi=0 \tag{4.1}$$

We can decompose the radial, angular, and temporal field components by setting:

$$\Phi=e^{-i\omega t}Y^l_m(\theta,\phi)R(r) \tag{4.2}$$

and then we can write the radial equation as:

$$\left[\frac{h(r)}{r^2}\frac{d}{dr}h(r)r^2\frac{d}{dr}+\omega^2-h(r)\left(\frac{l(l+1)}{r^2}+\mu^2\right)\right]R(r)=0. \tag{4.3}$$

Where $h(r)$ is defined by the Schwarzschild metric as:

$$ds^2=-h(r)dt^2+\frac{1}{h(r)}dr^2+r^2d\Omega^2. \tag{4.4}$$

Now we can make a transformation of coordinates, taking $r\to r_\star$ and $R(r)\to U(r)$ with $r_\star$ the tortoise coordinate, such that:

$$dr_\star=\frac{dr}{h(r)} \qquad\qquad U(r)=rR(r). \tag{4.5}$$

In these coordinates the radial equation assumes a Schrödinger-like form

$$\frac{d^2U}{dr_\star^2}+\Big(\omega^2-V_l(r)\Big)U=0, \tag{4.6}$$

with an l-dependent effective potential given by:

$$V_l(r)=\left(1-\frac{R_s}{r}\right)\left(\frac{l(l+1)}{r^2}+\frac{R_s}{r^3}+\mu^2\right). \tag{4.7}$$

We see that a local minimum, and hence a classically stable orbit, can appear for some values of μ and l. Below, we plot this effective potential for $\mu=.1*R_s^{-2}$ and $l=0,1,2$ and find that there is a stable orbit for $l=0$ along with unstable orbits for other parameter choices. We find these minima by taking $\frac{d}{dr}V_l=0$, from which we obtain:

$$R_s\mu^2r^3-2l(l+1)r^2-3R_s(1-l(l+1))r+4R_s^2=0. \tag{4.8}$$

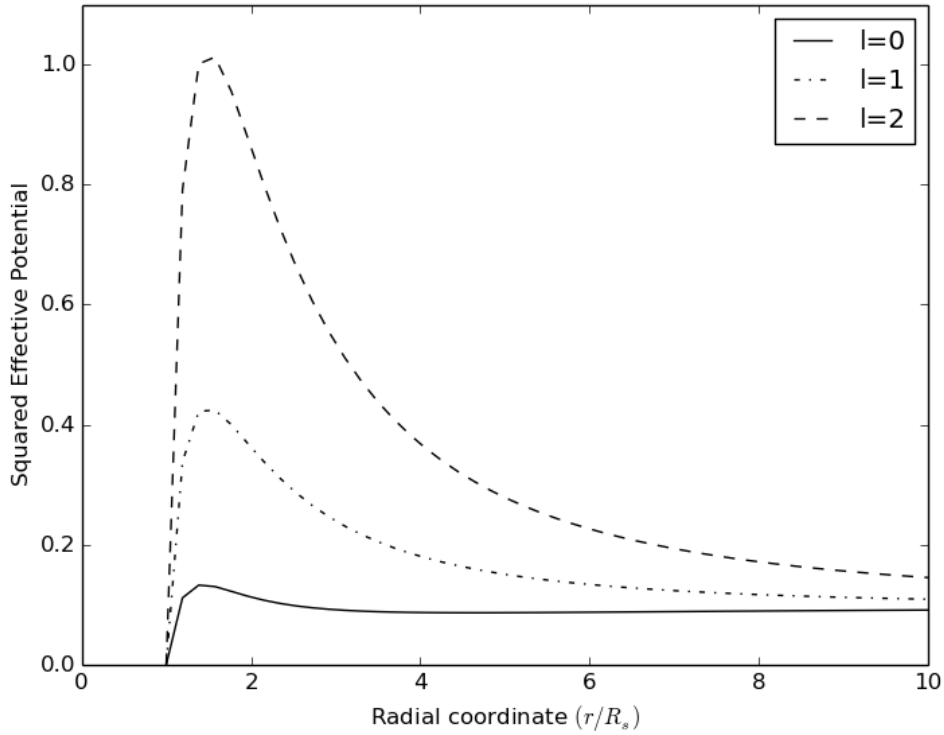

Figure 4.1: Effective potential for a Klein-Gordon particle on the Schwarzschild background.

We see that if the value of μ is less than some critical value, then there exist two roots outside the horizon $r > R_s$. This critical value for the mass of our Klein-Gordon particle is found to be:

$$\mu_{max} = \frac{1}{216R_s^2}\big(-27A + \sqrt{729A^2 + 432B}\big) \tag{4.9}$$

$$A = l^3(l+1)^3 + l^2(l+1)^2 - l(l+1) - 1 \tag{4.10}$$

$$B = l^2(l+1)^2[9 + 14l(l+1) + 9l^2(l+1)^2], \tag{4.11}$$

and for the special case of $l = 0$ we have $\mu_{max} = \frac{1}{2R_s}$. From the plot above we can see that there exists a clear minimum for the $l = 0$ case plotted, and in general a stable orbit for the $l = 0$ case if the mass of the Klein-Gordon particle is $< \frac{1}{2R_s}$. Moreover, for any particle with $\mu < \frac{1}{2R_s}$ there exists a local barrier and potential well (and hence a stable orbit) for all values of the angular momentum. There is more, very interesting, analysis that can be done on this effective potential and the conditions on angular momentum that arise if one desires bound states for $\mu > \frac{1}{2R_s}$, but we have

found both stable and unstable orbits in an effective potential being acted on by a Schrödinger-type operator. Is this enough to confirm the existence of bound states? Unfortunately not, and more analysis is needed. Even when there is a well in the potential, this does not *guarantee* bound states to exist, the well must be deep enough to ensure stability because of the possibility of tunnelling into the global minimum at the origin.

However, there is reason to believe that the culprit for our lack of bound states in other settings may be due to more than just the spin angular momentum. The Klein-Gordon equation has another property that distinguishes it from the Dirac case, it does not conserve probability. This prevents us from interpreting the associated wave-function Ψ as a probability amplitude. In order to support the interpretation of a probability amplitude for Ψ we need a probability density $\rho(r)$ and a current $\vec{J}(x)$ that satisfy a continuity equation

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0, \tag{4.12}$$

to ensure that no probability is 'lost.' It can be shown [15] that one *can* satisfy the continuity equation with a solution to the Klein-Gordon equation, but the density is not positive definite. This means that in order to conserve probability, the theory allows for negative probabilities, which disallows any obvious physical interpretation.

Perhaps then, the distinguishing feature that allows for bound states in the Klein-Gordon setting as compared to the Dirac equation is tied up in the probability current. Indeed, our result from Section 2.5 in which we presented a non-Hermitian Hamiltonian-type operator for the Dirac equation on the Schwarzschild background, and conjectured that the lack of Hermiticity was due to the singularity at the origin acting as a probability sink, serves as another motivation to investigate the Dirac current in a curved background.

4.2 Divergence of the Dirac Current Across the Event Horizon

Again we take the massive Dirac equation in the Schwarzschild background

$$[\gamma^a e_a^\mu (\partial_\mu + \Gamma_\mu) + \mu]\, \Psi = 0, \tag{4.13}$$

with $\mu = \frac{mc}{\hbar}$ and we decompose this in the normal way [6] into two coupled radial equations:

$$\left[e^{-\nu}(h-V)+\mu e^{-\nu/2}\right]F+\frac{dG}{dr}-\frac{k}{r}Ge^{-\nu/2}=0 \tag{4.14}$$

$$\left[e^{-\nu}(h-V)-\mu e^{-\nu/2}\right]G+\frac{dF}{dr}-\frac{k}{r}Fe^{-\nu/2}=0 \tag{4.15}$$

with $h = \frac{E}{\hbar c}$, $V = \frac{-qe}{\hbar cr}$, and k is the eigenvalue of the angular K operator. F and G are the components, up and down, respectively, of the two-component spinor that solves the radial equation. If we expand Eq. 4.14 around $r = 2M$, we find that the leading terms are:

$$(r-2M)F'+2MEG=0 \qquad\qquad (r-2M)G'-2MEF=0, \tag{4.16}$$

and the solutions for F and G are given by:

$$F(r), G(r) = \cosh\left(2ME\ln\left(r-2M\right)\right) \pm i\sinh\left(2ME\ln\left(r-2M\right)\right) \tag{4.17}$$

Now, for the Dirac current density we have:

$$J^0 = \bar{\Psi}\sqrt{-g}\gamma^0\Psi \approx \sqrt{-g}\frac{F^2+G^2}{r^2g^{00}} \tag{4.18}$$

and so the integral

$$\int J^0 dr d\theta d\phi = \int \sqrt{-g}\frac{F^2+G^2}{r^2g^{00}} dr d\theta d\phi \tag{4.19}$$

diverges at $r = 2M$, the radius of the event horizon. So our density becomes infinite and thereby our wave-function is not normalizable and hence has no bound states. While this method is certainly approximate, and the demonstration is only for the Schwarzschild background, it lends more credence to the conjecture that the culprit for the lack of bound states is that the presence of an event horizon disallows normalizable solutions. Indeed, this idea is supported by a paper by Cohen and Powers [9] wherein they rigorously demonstrate, through the use of spectral methods, that for Schwarzschild and non-extreme Reissner-Nordstrom geometries (that is, metrics with a clothed singularity) there exist no bound states, but in the naked singularity case bound states are allowed. As mentioned in Chapter 3, this result was extended to the extreme Kerr-Newman case by Schmid [21]. In the next section we will at-

tempt to probe the role of the event horizon further by adding making the extreme Kerr-Newman case more complex, and seeing if the introduction of another separate event horizon kills the bound states.

4.3 The Kerr-Newman-dS Black Hole

In this section we consider the geometry set up by a Kerr-Newman black hole in the presence of a positive cosmological constant. We can show that simply through the introduction of a cosmological constant into the *vacuum* Einstein equations one can introduce an event horizon into the metric. The vacuum Einstein equations with a positive cosmological constant are:

$$R_{\mu\nu} = \Lambda g_{\mu\nu}. \tag{4.20}$$

For which a spherically symmetric solution can be found, the de-Sitter metric:

$$ds^2 = -\left(1 - \frac{\Lambda r^2}{3}\right) dt^2 + \left(1 - \frac{\Lambda r^2}{3}\right)^{-1} dr^2 + r^2 \left(d\theta^2 + \sin^2\theta d\phi^2\right). \tag{4.21}$$

We can re-write this metric with the definition of a quantity $l \equiv \sqrt{\frac{\Lambda}{3}}$ and find

$$ds^2 = -\left(1 - \frac{r^2}{l^2}\right) dt^2 + \left(1 - \frac{r^2}{l^2}\right)^{-1} dr^2 + r^2 \left(d\theta^2 + \sin^2\theta d\phi^2\right), \tag{4.22}$$

where we see that for $r = l$ the co-efficient in front of dr^2 diverges, indicating the presence of an event horizon, often called the *cosmological horizon*. Re-inserting the presence of mass, charge, and angular momentum retains this cosmological horizon, while picking up the event horizons we are more familiar with. For the case of a Kerr-Newman black hole sitting in this de-Sitter universe we have three horizons: the black hole event horizon, a Cauchy horizon, and the cosmological horizon.

In the extreme Kerr-Newman setting the black hole horizon and Cauchy horizon become degenerate, and it has been shown that in this case there *do* exist bound states for the Dirac equation[21]. In order to support our hypothesis that the critical component in killing the bound states is an event horizon; the hope is that in the Kerr-Newman-dS case, even when the event horizon and Cauchy horizon become degenerate in the extremal setting, the presence of another horizon (*cosmological*) will kill off the bound states.

Our suspicion is vindicated. It has been shown by Belgiorno and Cacciatori [4] that this is the case. They use techniques that will be familiar to the reader from our non-existence result in Chapter 3. That is, they separate out a radial Hamiltonian for their system, take it to the tortoise-like coordinates, and for the non-extreme case they show that its spectrum is continuous in $\mathbb{R}$ except for a special point. As we did in Chapter 3, they then utilize the Levinson theorem to show that this special point is *not* an eigenvalue of the non-extreme Hamiltonian. They are then able to exploit the fact that for the Kerr-Newman-dS system, the extremal operator is unitarily equivalent to the non-extremal one, hence yielding the same spectrum.

Conclusion

When first learning elementary Quantum Mechanics, I was told not to worry about its counter-intuitive predictions, that my confusion was normal and natural, and that no one *really* understands it. However, after doing enough problems I began to form a new set of heuristics and intuitions for how quantum mechanical systems would work, as if learning the rules to a new game. Being exposed to what were really the same few problems with varying levels of sophistication tricked me into thinking I had spanned the space with these new intuitions. This thesis was an exciting and rude awakening as my assumptions were consistently subverted. I spent the entire year trying to chase a bound state, at every turn thinking that the introduction of some new machinery, or term or framework would provide me with one...to no avail. In writing this document I tried my best to replicate this experience for the reader: you are the donkey and the bound state is the carrot and the stick is the issue I am now obsessed with. I couldn't help but chase my curiosity down the rabbit hole, and it took me to a world where all my assumptions were disposed of and where the physics I desperately wished to understand was consistently out of reach.

I feel the need, but not the qualification, to comment on how I think the reader should take the results presented in this thesis. With Quantum Mechanics there is often a sense that one should be able to drop any potential in and have something come out, when in reality there are only two workable and physically realizable potentials. One could say that this story is just another such deficiency. Whether this deficiency is caused by the fact that we have worked in a single particle, and not field theoretic, framework is unclear and is a promising avenue for future research.

If there are any take-a-ways from this thesis they are as follows: the spectral theory is an exceptionally powerful tool, black holes are still stranger than fiction, and if you hear me starting to wax philosophical about the issues with naive theory combination, run away.

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